NAZAR VISION

Entity relationship model

Hatchling detection proof of concept | Data lineage, annotation state, and experiment artifacts

PROPOSED DESIGN

Version 1.0 | 18 Oct 2026

DATA LIFECYCLE



The arrows indicate lineage through frozen data and saved artifacts, not live database joins.

MODEL BOUNDARY

One persistent FiftyOne working dataset, backed by MongoDB, owns the samples, tags, and annotation/prediction fields.
Logical boxes in this ERD may be embedded values, configuration entries, or versioned files; no SQL tables are implied.

HOW TO READ THE DIAGRAMS

1 = exactly one 0..* = zero or many 1..* = one or many

PK and FK denote logical identity and lineage, not relational constraints.

Arrows point from parent or input to child or produced artifact.

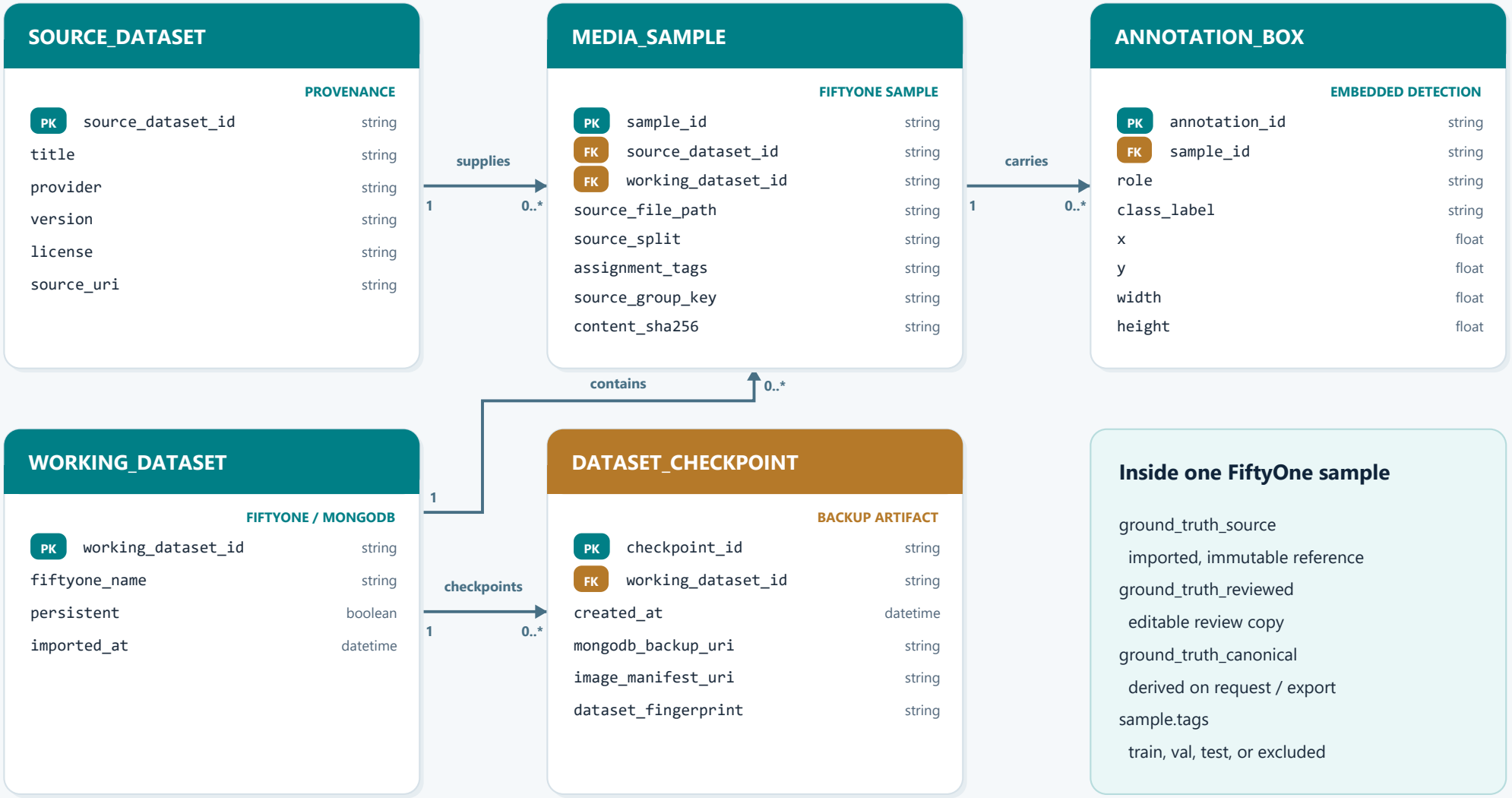
02 Dataset & annotation 03 Export & training 04 Evaluation 05 Video

06 Data invariants 07 Storage realization

Source: Nazar Vision project brief, especially pp. 5-16 and 25-27.

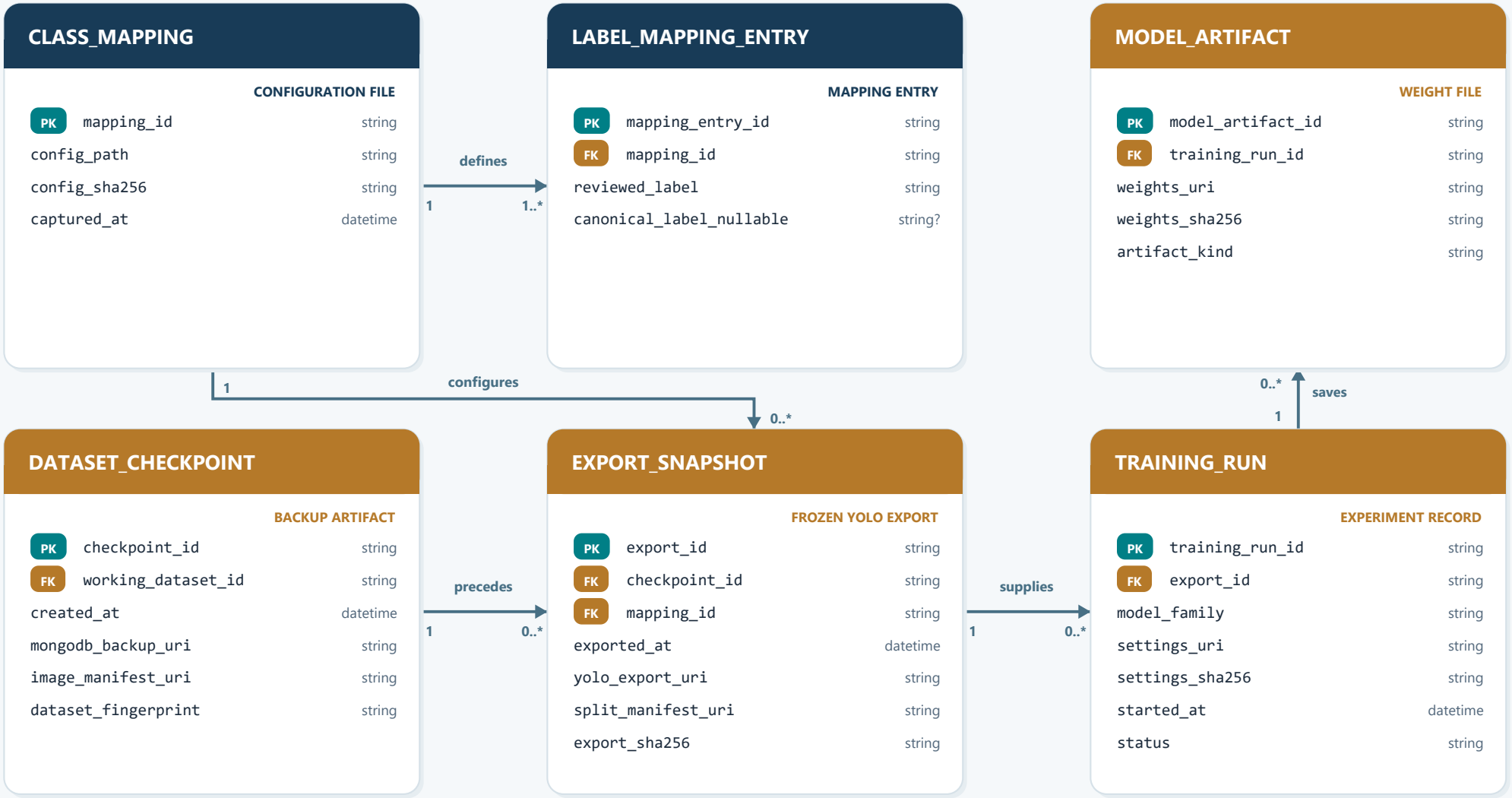
Dataset and annotation

Source provenance and review state live within the persistent FiftyOne working dataset.



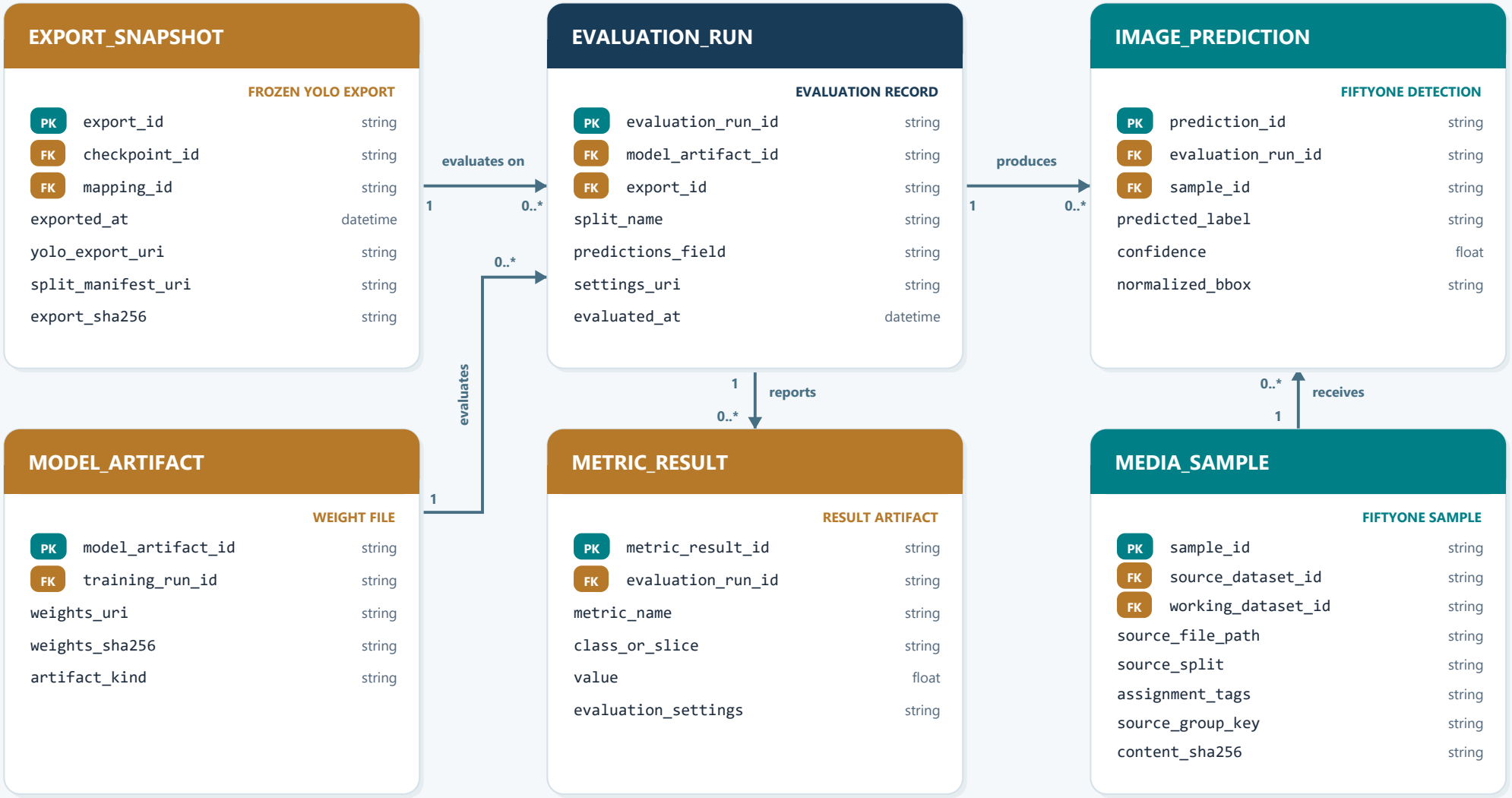
Mapping, export and training

The frozen export records which checkpoint and class mapping produced each training input.



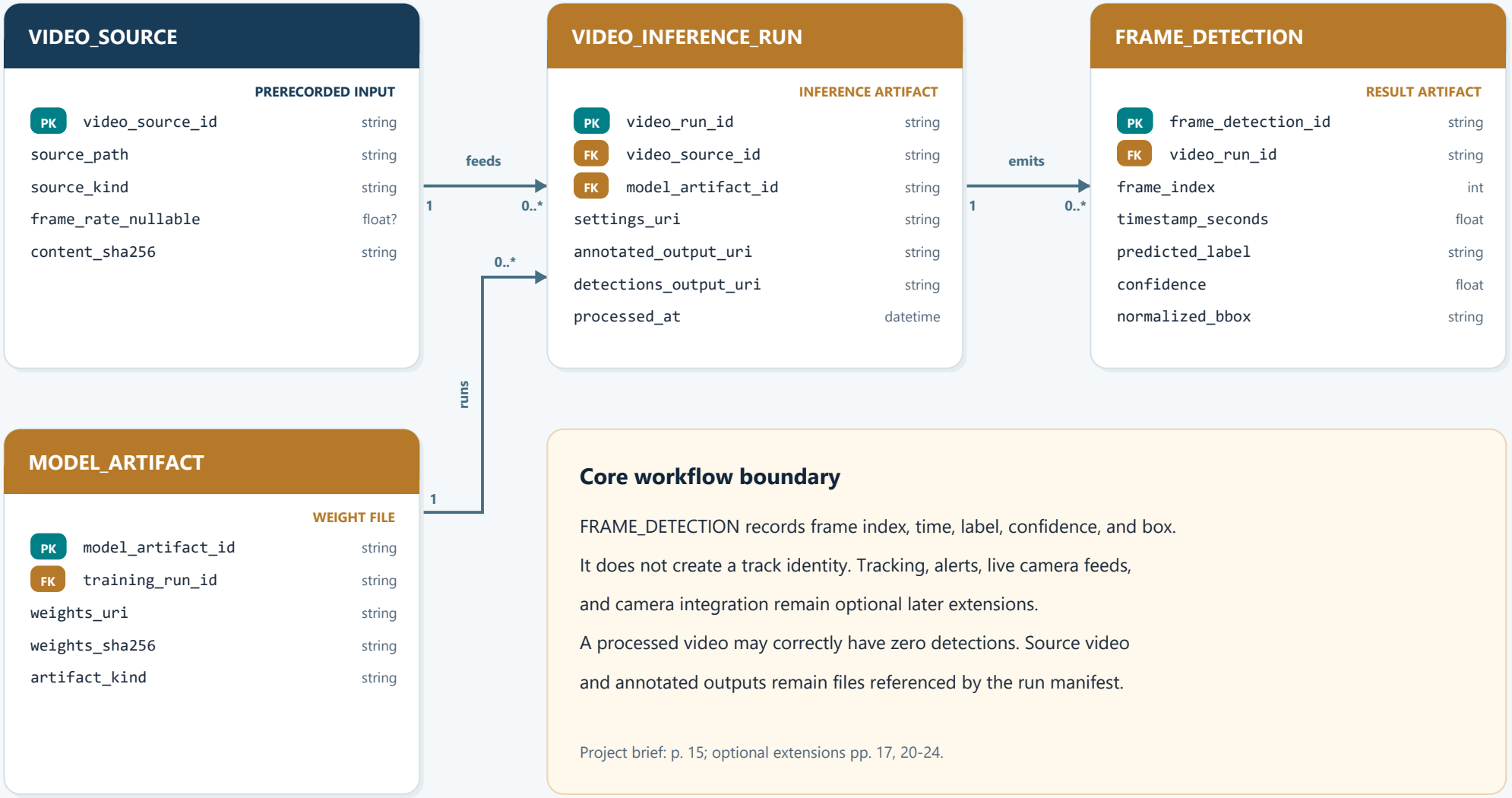
Model evaluation and predictions

Each evaluation chooses one saved weight file and one frozen export; its predictions stay isolated by run.



Prerecorded video inference

A saved weight file runs on a video or ordered image sequence and produces frame-level detections.



Core workflow boundary

FRAME_DETECTION records frame index, time, label, confidence, and box.

It does not create a track identity. Tracking, alerts, live camera feeds,

and camera integration remain optional later extensions.

A processed video may correctly have zero detections. Source video

and annotated outputs remain files referenced by the run manifest.

Project brief: p. 15; optional extensions pp. 17, 20-24.

Data invariants and validation gates

These conditions make the logical model useful for reproducible training and honest evaluation.

01 — Preserve originals

Source images and imported boxes remain immutable. Reviewed boxes are a separate editable field.

Brief pp. 7-10

02 — Persist the working dataset

FiftyOne dataset.persistent is true; reopened state retains boxes, tags, canonical labels, and metadata.

Brief pp. 7, 12-13

03 — Assign one effective split

Exactly one train, val, test, or excluded tag applies at export; conflicting or missing tags block export.

Brief pp. 9, 12

04 — Derive canonical labels

Generate canonical from reviewed boxes and mapping. Unknown reviewed labels cause an explicit error.

Brief pp. 10-11

05 — Retain background images

Mapping a non-target label to null removes its box, not the image or its valid split assignment.

Brief pp. 6, 10-11

06 — Validate export boxes

Require expected class, finite normalized coordinates, positive size, and bounds within the image.

Brief p. 12

07 — Investigate leakage

Flag duplicate and related imagery across splits; record unresolved sequence or augmentation risks.

Brief pp. 6, 12

08 — Bind recoverable inputs

Checkpoint before export; save mapping, split manifest, settings, weights, and tested image-path recovery.

Brief pp. 12-14, 26

09 — Isolate predictions

Use a unique FiftyOne prediction field per evaluation; never overwrite any ground-truth field.

Brief pp. 14-15

10 — Keep final test independent

Use train/validation for development choices; record split, thresholds, metric definitions, and model version.

Brief pp. 14-15, 26

Logical-to-physical storage mapping

Logical IDs support traceability; storage uses FiftyOne/MongoDB, configuration, and versioned artifacts.

LOGICAL CONCERN	PHYSICAL REALIZATION	DESIGN CONSEQUENCE
Source image + provenance	Read-only image path on FiftyOne sample; source/import manifest and metadata.	MongoDB backup omits image bytes; restore needs files or path remap.
Working annotations + tags	One persistent FiftyOne dataset; three ground-truth fields and sample.tags.	Review state has one authoritative home in the working dataset.
Class mapping	Versioned project configuration file; hash and entries captured at export.	Every export is tied to the exact label transformation used.
Checkpoint + YOLO export	MongoDB backup and image manifest; frozen images/labels, data.yaml, splits.	A run remains reproducible after later FiftyOne review edits.
Training + image evaluation	Ultralytics run/weight files; unique FiftyOne prediction field per run.	Predictions can be reviewed beside ground truth without overwriting it.
Video inference	Source video/frames, annotated output, optional detections JSON/CSV.	The core demo uses saved weights and needs no tracking identity.

Design status: names for IDs, hashes and manifests are proposed metadata; reconcile them with the mentor's existing codebase.